

5. A 25.0 cm³ sample of a solution containing iodine is titrated against a sodium thiosulfate solution of concentration 0.0200 mol dm⁻³. This requires 23.25 cm³ of sodium thiosulfate for complete reaction.



Calculate the concentration of the iodine in the solution in g dm⁻³.

[2]

Concentration = g dm⁻³

6. The enthalpy of solution of sodium chloride is +4 kJ mol⁻¹. Explain why this compound is soluble in water despite this value being positive. [1]

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7. When a solution containing the weak base ammonia is neutralised using the strong acid sulfuric acid, a solution of ammonium sulfate is formed. Suggest a pH for this solution, giving a reason for your answer. [2]

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5. When CCl_4 is added to water two separate layers are formed and there is no reaction, whilst SiCl_4 reacts violently with water.

Give the reason for this difference.

[1]

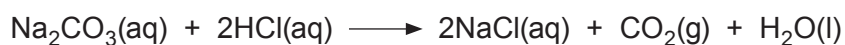
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6. The entropy change of the reaction shown below is positive. Give a reason for this.

[1]



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7. Carbon monoxide can be used to extract iron by the reduction of Fe_3O_4 .

(a) Write an equation for this reaction.

[1]

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(b) Give a reason why carbon monoxide acts as a reducing agent while the equivalent oxide of lead does not.

[1]

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8. Draw the **structure** of the copper-containing ion formed when concentrated hydrochloric acid is added to a solution containing $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ ions.

[1]



5. Phosphorus is able to form two different chlorides, PCl_3 and PCl_5 . Nitrogen is only able to form one chloride, NCl_3 .

Explain this difference.

[1]

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6. Write the expression for the ionic product of water, K_w .

[1]

7. Give a reason why the entropy of Hg(l) is greater than that of Au(s) .

[1]

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5. Addition of ammonia solution to aqueous copper(II) sulfate causes a pale blue precipitate to form followed by a coloured solution when excess ammonia solution is added.

Give the colour of the solution formed.

[1]

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6. The Haber process uses a heterogeneous catalyst.
State what is meant by the term 'heterogeneous' in this context.

[1]

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7. Hexagonal boron nitride is sometimes called white graphite. Give **one** difference between the physical properties of hexagonal boron nitride and graphite.

[1]

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